Research Question and Hypothesis (Ava)

We chose to investigate marine food webs, which are composed of predator-prey or feeding relationships in aquatic environments. Our group will be tackling the following research questions: How does community composition shape the creation and persistence of a marine ecosystem's food web? Are species higher up in the food chain more connected and/or exhibit greater diet diversity than primary producers and low-level consumers? Regarding the first question, we hypothesize that a range of primary producers, primary consumers, secondary consumers, tertiary consumers, and decomposers are required to create a robust food web within a marine ecosystem. Note that within our dataset, we sort organisms based upon trophic levels. Considering the second question, we hypothesize upper-level consumers will be highly connected species and exhibit greater diet diversity compared to primary producers and low-level consumers because they feed on a range of energy sources. Accordingly, we predict upper-level consumers will be characterized by many connections within the network, measured by their node possessing a high degree. Marine environments constitute the majority of Earth's surface, perform vital ecosystem services, and serve as a food source for humans and other organisms. Safeguarding marine ecosystems involves analyzing their health to create informed, functional policies for appropriate, sustained human use. One indicator of ecosystem health is the persistence of a diverse, robust food web. Our goal for this project is to evaluate feeding relationships among marine organisms to construct a food web, analyze connectedness, and utilize network characteristics as indicators of ecosystem health and resilience.

Dataset (Robbie)

The dataset is "Torres, M. A. et al. Food-web structure of and fishing impacts on the Gulf of Cadiz ecosystem (Southwestern Spain) Ecological Modelling," made available at https://www.web-of-life.es/map.php?type=7 through The Web of Life. The data details the Gulf of Cadiz food web, where each edge is a weighted (0 to 1) directed edge indicating what share of the source node's food consumption comes from the target/recipient node (ex., if a shark's diet is 50% Tuna, the value of shark --> tuna = 0.5). The dataset encompasses 43* functional groups and 411* nonzero interactions.

* After data cleaning.

The original source was Torres et al. 2013, from the Spanish Institute of Oceanography (IEO) in Cádiz. The csv file containing the data used in our interpretation comes from the Web of Life page: https://www.web-of-life.es/map.php?type=7.

References for the literature section:

Torres, M. Á., Coll, M., Heymans, J. J., Christensen, V., & Sobrino, I. (2013). Food-web structure of and fishing impacts on the Gulf of Cadiz ecosystem (South-western Spain). *Ecological Modelling*, 265, 26–44. https://doi.org/10.1016/j.ecolmodel.2013.05.019

Fortuna, M. A., Ortega, R., & Bascompte, J. (2014). The Web of Life. *arXiv*:1403.2575. https://doi.org/10.48550/arXiv.1403.2575

The dataset mostly supports the first question: "Does community composition shape the web's creation and persistence?" It can help answer this question in that the data lists 43 groups (and the cleaned group of inputs) along with what they consume. If we remove certain groups or levels, we can then see which consumers are left with something to eat (or what percentage of their diet) and work from there... we do not, however, have a way to account for any other changes that might occur as a result of the removal in question.

The second question can be mostly answered with our dataset: "Are higher consumers more connected, with more diverse diets?" It can be answered because the dataset directly lists each group's weighted diet. Assuming we can find or calculate each group's trophic level, we can easily determine connectivity based on trophic level and answer the question. As stated, the only bottleneck here is determining trophic level, but that can be done with the data given (for example with the formula $TL_i = 1 + \Sigma_j (D_{ij} \times TL_j)$ where i is predator and j is prey).

Yes, the dataset has been downloaded (as a CSV file) from the Web of Life site. The download contains a README and two CSV files containing the full interaction data and general information. The main csv file contains a 44 x 44 matrix with 413 nonzero values (edges). It had no row or column values, so those were added from the Web of Life species list. Also, one node/group, "Import," was removed as it is not a real group but rather food eaten outside the study zone and brought in (for example, in a turtle's stomach). This is why the listed group count is 43 (containing 411 directed weighted edges). Also, trophic level needs to be calculated and determined for each group and added to the data. This does not go in the interaction matrix, but rather in a separate file for later trophic-level-based interpretation methods. This is calculated with the aforementioned equation.

The only variable is diet share for each predator – prey group. These values range from 0 – 1 and include direct interaction (self loops) indicating cannibalism. There is further information beyond the 44 x 44 matrix (cleaned to 43 x 43) in the README and additional info in the CSV (which includes info like location and connectivity). Network properties such as isolates (0), self loops (14), nodes (43), and edges (411) can be easily calculated. The trophic level can be calculated with the previously mentioned equation. Doing this for all groups and then modelling with trophic level as the y-axis and drawing directed nodes based on consumption produces the network detailed in Figure 1 below.

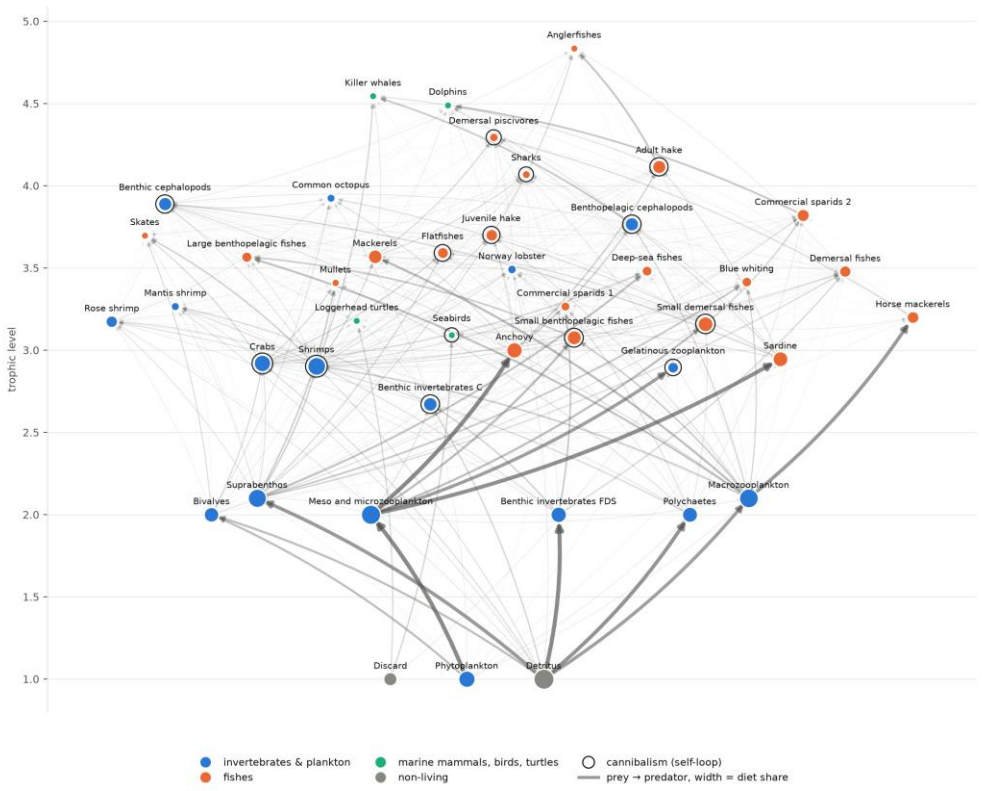


Figure 1. Group trophic level, broad feeding type, and directed & weighted nodes based on consumption indicated in the dataset.

Planned Methods (Emma)

We plan on creating a **weighted directed network** based on the aforementioned dataset that reflects the relative frequencies of predator-prey feeding. The directionality of our network reflects the one-sidedness of predator-prey relationships, in that predators consume prey but not vice versa. Weighted edges will help to represent the frequency of the prey species in the predator's diet. Synthesizing information about each species' trophic level, we can conclude whether species higher in the food chain exhibit greater diet diversity through a comparison of degrees. We will create a **degree distribution** to easily visualize the differences in degrees across species. Calculating the **clustering coefficient** would shed light on the robustness of the ecosystem and its ability to persist beyond disturbances as it reflects the connectedness of a node's neighbors, which would imply in our network that a species has a wider array of prey options and may be able to withstand a change in a single prey species' levels more successfully.

Literature (Lilly)

Frelat, R., Kortsch, S., Kröncke, I., Neumann, H., Nordström, M. C., Olivier, P. E. N., & Sell, A. F. (2022). Food web structure and community composition: A comparison across space and time in the North Sea. *Ecography, 2022*(2). https://doi.org/10.1111/ecog.05945

- This paper specifically studies how community composition changes food web structures over time and climate and found that community composition can be affected by climate, leading to effects in food web structures, directly addressing our research question just at a larger scale.

Endrédi, A., Patonai, K., Podani, J., Libralato, S., & Jordán, F. (2021). Who Is Where in Marine Food Webs? A Trait-Based Analysis of Network Positions. *Frontiers in Marine Science, 8*, 636042. https://doi.org/10.3389/fmars.2021.636042

- This paper looks at how trophic level and other species traits affect network connectivity interactions between species, which can help us to understand whether higher-level consumers are more connected within marine food webs.

Rooney, N., McCann, K. S., & Moore, J. C. (2012). Persistence increases with diversity and connectance in trophic metacommunities. *PLoS ONE, 7*(1), e19374. https://doi.org/10.1371/journal.pone.0019374

- This paper studies the complexity of food webs and how it can influence their persistence and found that increased complexity leads to persistence. This can help us to interpret our data and understand that if we observe a less complex food web, the likelihood of it persisting would be lower compared to if it had increased complexity.

Xu, Y., Peng, J., Qu, J., Cai, Y., Yang, Z., & Sun, X. (2020). Assessing food web health with network topology and stability analysis in aquatic ecosystem. *Ecological Indicators, 109*, 105820. https://doi.org/10.1016/j.ecolind.2019.105820

- This paper can inform us about an ecosystem's stability and health by highlighting food-web characteristics such as connectivity and how they correlate with ecosystem health. They specifically found that highly connected nodes are one characteristic of a healthy ecosystem, which can allow us to use node degrees to identify important measures of a food web.

Timeline (Lilly)

October 1 – Project Proposal

October 8 – Draft of Introduction

October 12 – Understand and process all the data

October 19 – Sort through the data to create our weighted network and complete initial data analyses to answer our research question

October 22 – Draft of Methods

November 5 – Draft of Results

November 12 – Draft of title, abstract and conclusion

November 19 – Final data analyses complete (finalized network and calculations that answer our research question)

November 30 – Final figures made and final paper written

December 1-2 – Presentation preparation

December 3 – Presentation Day

December 8 – Revised Draft